

Fault Intelligence as Code

AI Agents, RAG, and MCP for Network Operations

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Cisco Webex App

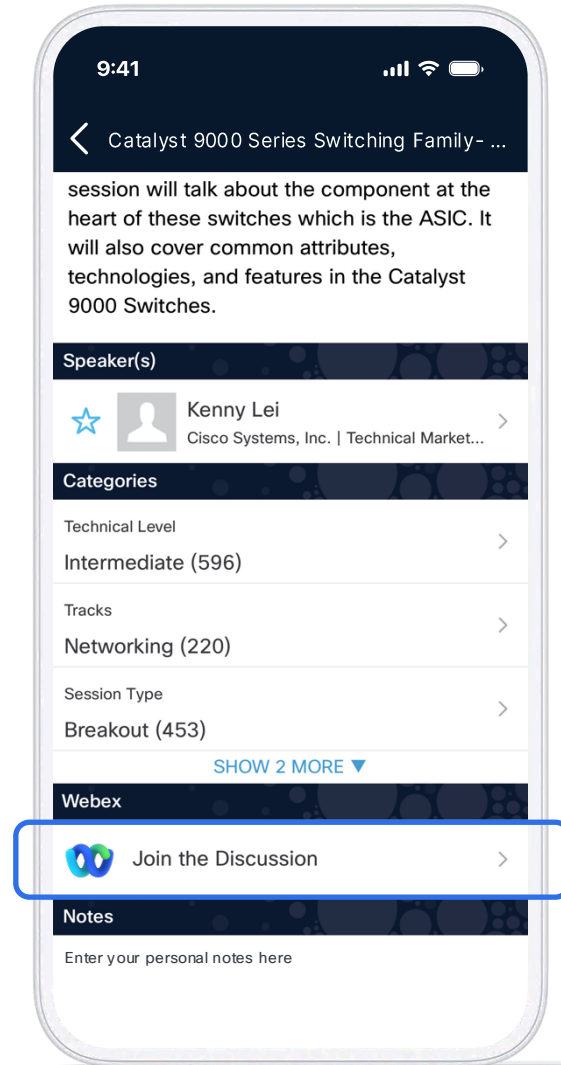
Questions?

Use Cisco Webex App to chat with the speaker after the session

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- 1 Find this session in the Cisco Live Mobile App
- 2 Click “Join the Discussion”
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Webex spaces will be moderated by the speaker until June 13, 2026.



<https://ciscolive.ciscoevents.com/clamer26/>
SESSIONCODE

Slides are available here



<http://cs.co/faut-intel-pdf>

About your speakers



Jason Shoemaker

- 14+ years architecting solutions for Web Hyperscale companies
- DC Networking → Network Automation → Reliability Engineering & AIOps
- Founder of CX Innovation Edge customer co-development initiative

Fun fact: Born the same year that Cisco was founded as a company



Steve Holl

- 20+ years at Cisco supporting the customer experience in Collaboration and Assurance
- TAC Customer Support → Managed Services → Assurance & Automation → Cisco IQ
- Cisco Press author, co-inventor of 7 patents and 6 pending

Fun fact: I never signed up for linkedin, but you can follow me on Strava instead, I guess!. 😊

A woman with short dark hair and glasses is looking at a screen in a server room. The background is filled with server racks and glowing green lights.

**Today's operational intelligence is
fragmented across staffing, tools, and
processes**

**And yet we still expect consistent, fast,
intelligent results? Every time?**

Agenda

- 01 Fault management challenges**
- 02 Fault Intelligence as Code principles**
- 03 Agentic network operations**
- 04 Fault Intelligence in action**
- 05 Try it yourself!**

Cisco's enterprise network assurance journey

1 Proactive monitoring (syslog, SNMP polling & traps, REST, etc.)

2 Methods of Procedure (Built a knowledge base, applied governance)

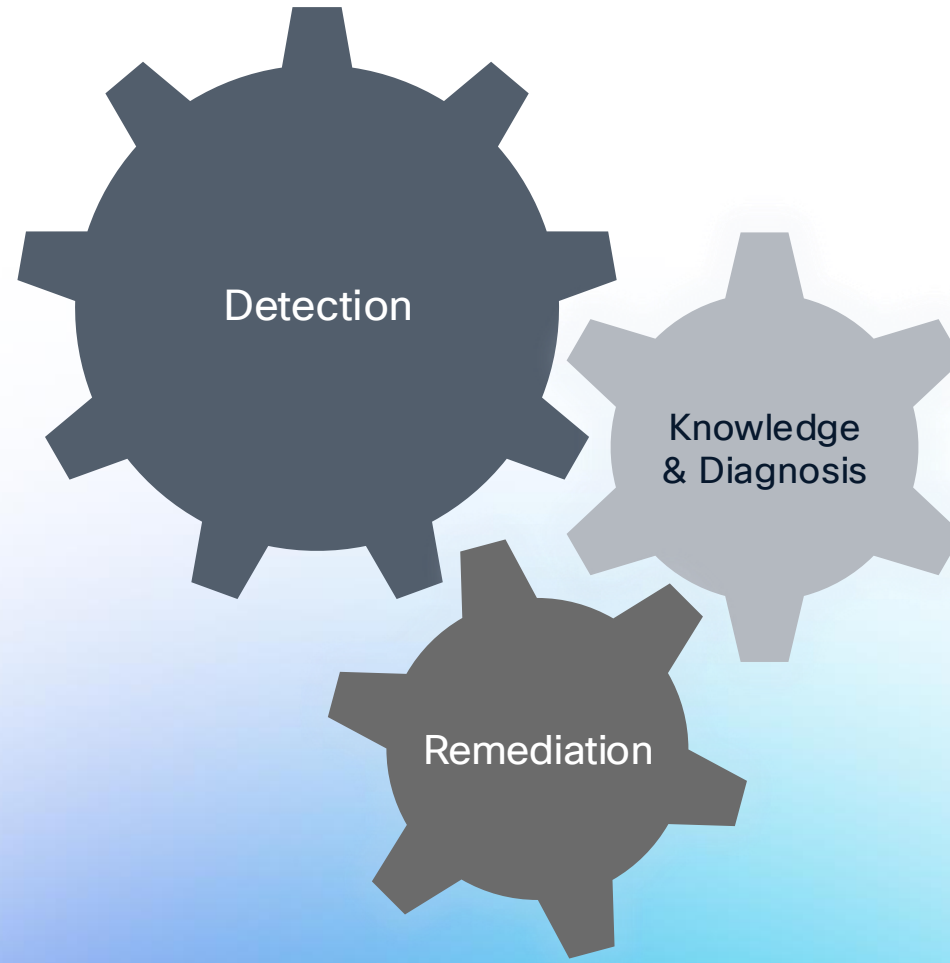
3 Synthetic User Testing/Monitoring (ThousandEyes)

4 Rule-based automated response
(fault triggering static workflow)

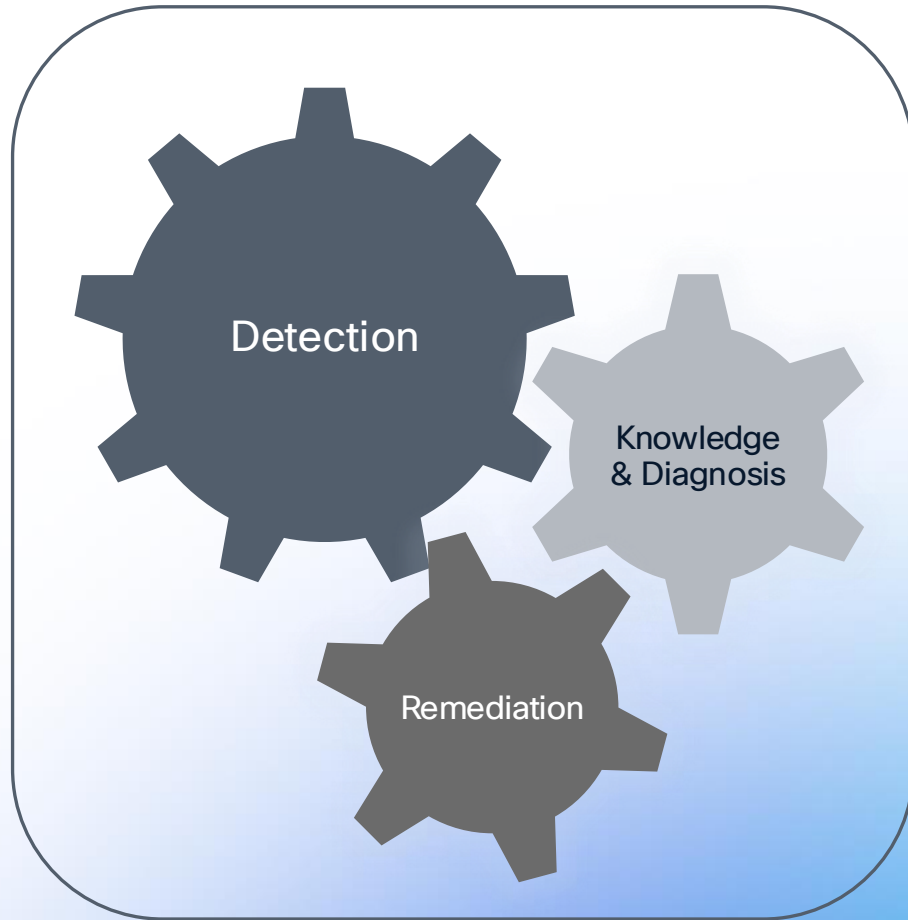
5 Provisioning via infrastructure as code
(Config in Git, implemented via CI/CD)

6 *What's next ???*

What if operational fault and incident knowledge was always-on, and always executing programmatically?



What if operational fault and incident knowledge was always-on, and always executing programmatically?



Provides context for...



Workflow-based automation



Agentic / AI-powered response

Managing Network Faults as Code



**We use CI/CD to deploy our infrastructure.
We use version control to manage configuration change.**

Now we can manage our faults and remediation practice as code, too.

What you will walk away with



**Knowing how to use
coding principles for
fault management**



**See how to integrate
fault intelligence-as-code
with agentic operations**

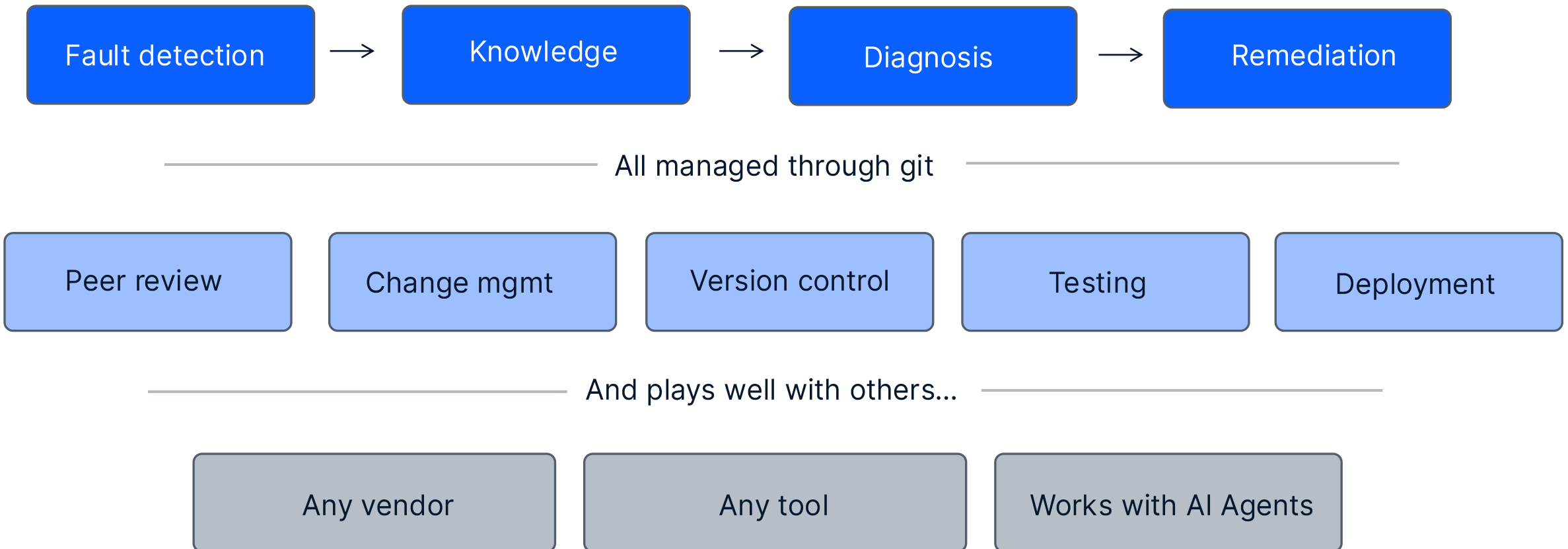


**Open-sourced code that
you can run for your
operations**



The principles of Fault Intelligence as Code

What it means to manage faults like code



What does fault intelligence as code look like?

Intelligence Artifacts (IA). Authoritative ops intelligence in YAML.

Fault Signature (FS) | Repair Action Workflow (RAW)

Data models. Machine-readable, structured.

Fault detection | Troubleshooting | Remediation

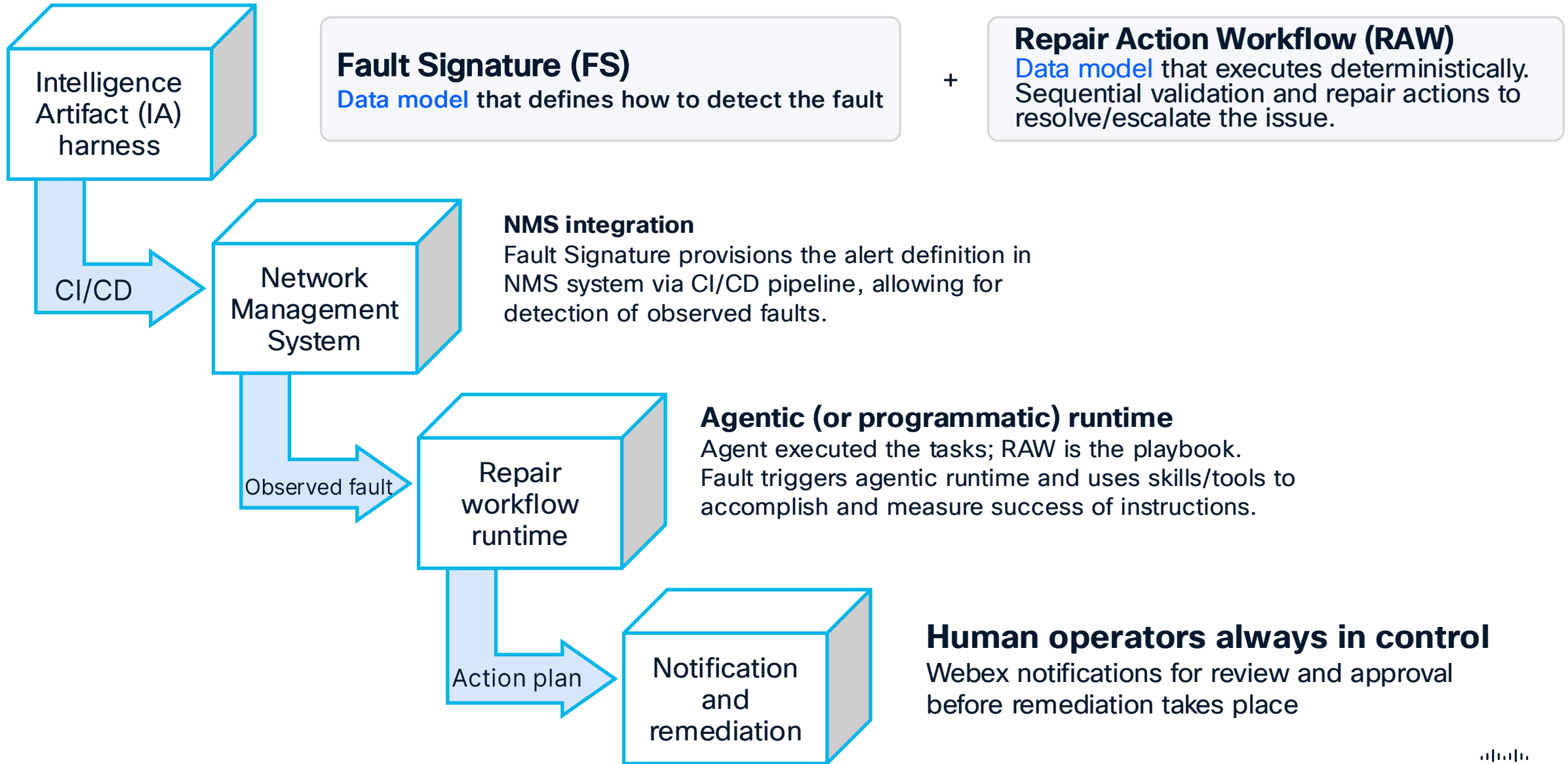
DevOps pipeline & discipline. Versioned, tested, deployed.

Continuously implemented and delivered. Leveraged by humans or agents.

Personalized. Curated from [your](#) operational data.

Understands your: Best-practices | Runbooks | Historical incident data

The functional components



Fault Signature (FS) schema

Overview

Provides a structured definition for monitoring system health.

Defines what events to match, the logic required to evaluate them, and the specific conditions under which an alert clears.

Used to provision fault alerting in a NMS.

Metadata

name, id, severity, component, product_ids, os_versions, tags

Event

Event types (syslog, alarm, metric, state, counter)
Event match criteria
Parameters to extract

Logic conditions

Boolean expression over events

clear_event

Pattern to identify the resolution of the fault

Fault Signature (FS) schema

yaml - BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED

```
schema_version: "0.0.1"

metadata:
  name: BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED
  id: "FS000003"
  alert_def_id: "AD000003"
  version: "1.0.0"
  description: |
    Detects an IOS XR BGP neighbor session that is
    torn down by the remote peer
  severity: WARNING
  created_date: "2026-05-29"
  modified_date: "2026-05-29"
  author: "CX Intelligence Team"
  product_ids:
    - "ASR 9000.*"
  os_versions:
    - "IOS.XR .*"
  component: BGP
  tags:
    - [bgp, routing, max-prefix, ios-xr, devnet-3171]
  priority: 5
  symptom: "LINK_DOWN"
  repair_action_workflow_ref:
    BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED_REPAIR
```

```
conditions:
  logic: "E1"
  logic_lookback_time: 0
  events:
    - event:
        id: "E1"
        type: syslog
        path: "syslog"
        message_type: "ROUTING-BGP-5-ADJCHANGE"
        message_sample: "bgp[1090]: %ROUTING-BGP-5-ADJCHANGE : neighbor 172.20.20.17
        Down - BGP Notification received, maximum number of prefixes reached (VRF: default)
        (AS: 65535)"
        evaluation:
          type: regex
          value: "%ROUTING-BGP-5-ADJCHANGE\\s*:\\s*neighbor\\s+(\\S+)\\s+Down\\s+
          \\s+BGP Notification received,\\s+maximum number of prefixes
          reached\\s+\\(VRF:\\s*(\\S+)\\)\\s+\\(AS:\\s*(\\d+)\\)"
          parameters:
            - type: extract_to_variable
              name: neighbor_ip
              source: "match.group(1)"
              description: "IP address of the affected BGP neighbor"
          match_count: 1
          match_period: 0
          clear_event:
            pattern: "%ROUTING-BGP-5-ADJCHANGE\\s*:\\s*neighbor\\s+{{ neighbor_ip
            }}\\s+Up\\s+\\(VRF:\\s*{{ vrf_name }}\\)\\s+\\(AS:\\s*{{ neighbor_as }}\\)"
            lookback_period: 300
          correlation_variables:
            - neighbor_ip
            - vrf_name
            - neighbor_as
```

Fault Signature as Code

Pushing a fault signature to a network management system

yaml - BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED

```
schema_version: "0.0.1"

metadata:
  name: BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED
  id: "FS000003"
  alert_def_id: "AD000003"
  version: "1.0.0"
  description: |
    Detects an IOS XR BGP neighbor session that is
    torn down by the remote peer
  severity: WARNING
  created_date: "2026-05-29"
  modified_date: "2026-05-29"
  author: "CX Intelligence Team"
  product_ids:
    - "ASR 9000.*"
  os_versions:
    - "IOS.XR .*"
  component: BGP
  tags:
    - [bgp, routing, max-prefix, ios-xr, devnet-3171]
  priority: 5
  symptom: "LINK_DOWN"
  repair_action_workflow_ref:
    BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED_REPAIR
```

CI/CD
provisions

SPL - SPLUNK SAVED SEARCH

Settings

Alert ad000003_bgp_neighbor_max_prefix_limit_exceeded

Description

Detects an IOS XR BGP neighbor session that is torn down by the remote peer because the local router advertised more prefixes than the peer's configured

Search

```
index=syslog sourcetype=cisco:ios "ROUTING-BGP-5-ADJCHANGE"
| regex _raw="%ROUTING-BGP-5-ADJCHANGE\s*:\s*neighbor\s+(\S+)\s+Down\s+-\s+
+BGP Notification received,\s+maximum number of prefixes reached\s+\(VRF
:\s*(\S+)\)\s+\(AS:\s*(\d+)\)\s+"
| rex field=_raw "%ROUTING-BGP-5-ADJCHANGE\s*:\s*neighbor\s+(?<neighbor_ip>\S+)\s+
+Down\s+-\s+BGP Notification received,\s+maximum number of prefixes reached\s+
+\(VRF:\s*(?<vrf_name>\S+)\)\s+\(AS:\s*(?<neighbor_as>\d+)\)\s+"
| eval alert_def_id="AD000003"
| table _time device_name device_ip alert_def_id neighbor_ip vrf_name neighbor_as
_raw
```

Trigger Actions

+ Add Actions ▾

When triggered

>	 Add to Triggered Alerts	Remove
▽	 Webhook	Remove
	URL http://198.18.128.50:8080/fault-alert	
	Specified URL to send JSON payload via HTTP POST (ex., https://your.server.com/api/v1/webhook).	
	Learn More	

Repair Action Workflow (RAW) schema

Overview

Instructions for "How do I fix it?"

Portable, ordered procedure that runs on a Fault Management System or agent.

Metadata

Name, version of workflow

Workflow steps

1. Each step includes evaluation step(s) and action selection.
2. Action selection evaluates conditions against validation outputs.

Validation actions

`eval_cli`, `eval_logs`, `eval_var`, conditional logic (and/or)

Repair actions

`exec_cli`, `config_cli`, `custom_action`, `wait`

Terminal actions & control flow

Terminal: `resolve`, `escalate`, `fail`
Control: `goto`, `revalidate`, `wait`

Repair Action Workflow (RAW)

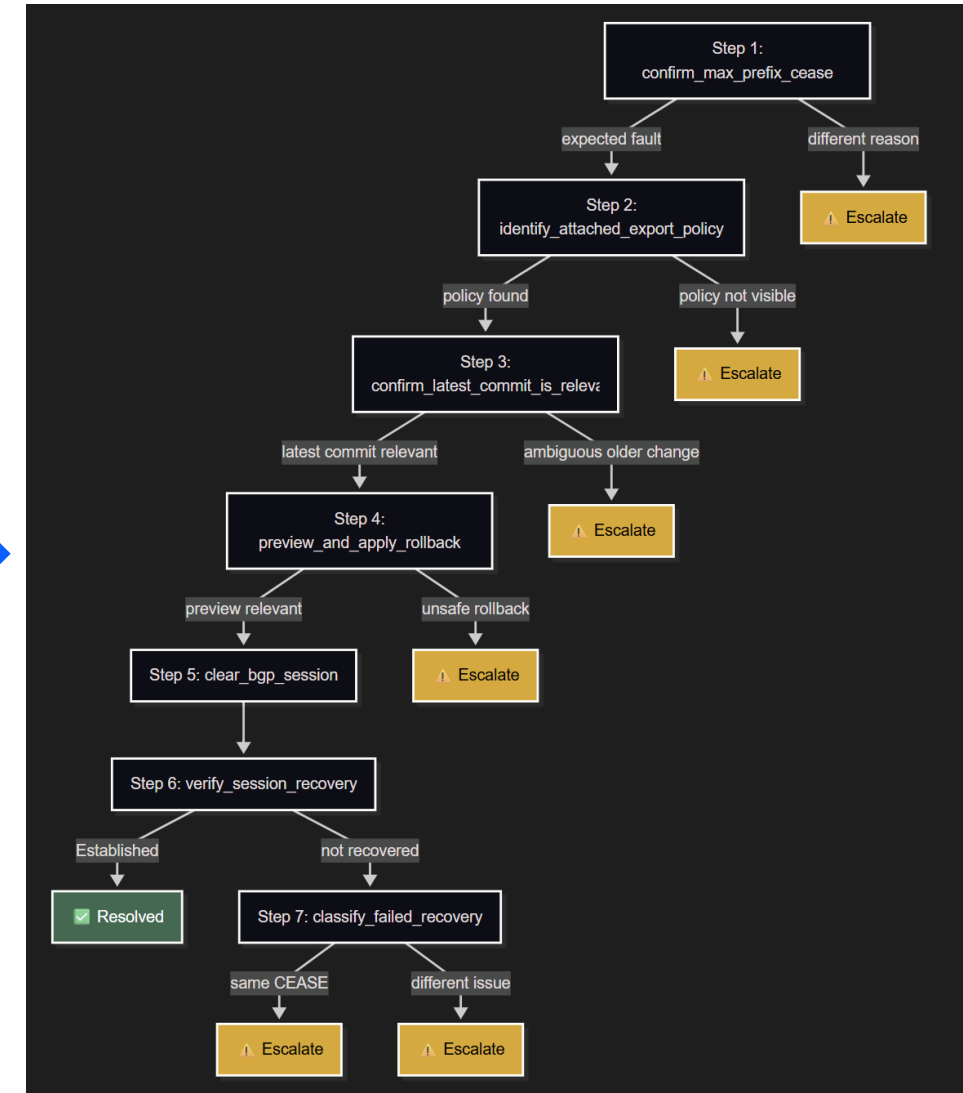
BGP_MAX_PREFIX_WORKFLOW

```
inputs:
- name: neighbor_ip
  source: "{{ alert_vars.neighbor_ip }}"
  type: string
  description: "IP address of the affected BGP neighbor extracted by FS000003"
- name: vrf_name
  source: "{{ alert_vars.vrf_name }}"
  type: string
  description: "VRF name extracted by FS000003"
...

action_groups: []

steps:
- step_id: "1"
  name: confirm_max_prefix_cease
  description: |
    Confirm that the BGP session is down due to a neighbor-sent max-prefix
    CEASE notification, not a different adjacency failure.
  validation:
    and:
      - eval_cli:
          commands:
            - "show bgp neighbors {{ neighbor_ip }} | include \"BGP state\""
          pattern: "BGP state = (Idle|Active)"
          outputs:
            - name: session_down
              source: "{{ result.matched }}"
              type: boolean
              description: "True when the neighbor is not in Established state"
  action_select:
    - action_id: "1a"
      name: fault_confirmed
      description: "Session is down and the reset reason matches the expected max-prefix CEASE."
      conditions:
        - condition: "session_down == True and max_prefix_cease_detected == True"
      action_groups:
        - actions:
            - goto:
                step_id: "2"
```

Drives
deterministic
action



Now we have some fundamental knowledge...



Knowledge of data models for operational fault management



See how to integrate fault intelligence-as-code with agentic operations



Open-sourced code that you can run for your operations



Fault Intelligence as Code for Agentic Network Operations

AI terms we'll use today

Term	What is it?
Agent	Agent = Harness + Model. Reasons, uses tools, follows instructions, maintains context, and take actions toward a goal
Agent Harness	The runtime environment around an LLM that makes it useful as an agent: prompts, tools, skills, memory, context management, execution loops, and safety controls
Context Engineering	The practice of giving an agent the right information, in the right structure, at the right time so it can perform a task reliably
Agent Skills	Reusable, task-specific instructions, procedures, and tools that an agent loads when needed to perform specialized work
MCP	Model Context Protocol: a standard way to expose external tools, APIs, and systems to an agent through a consistent interface
RAG	Retrieval Augmented Generation: retrieving information from external sources and adding it to the agent's context so it can generate a more accurate, grounded response

Agent harness design for rapid prototyping

Agent harnesses are great for writing code... *but what about troubleshooting network faults?*

Agent harness SDKs let you run coding agent harnesses programmatically

Use the same agent harness through an API for repeatable, automated execution

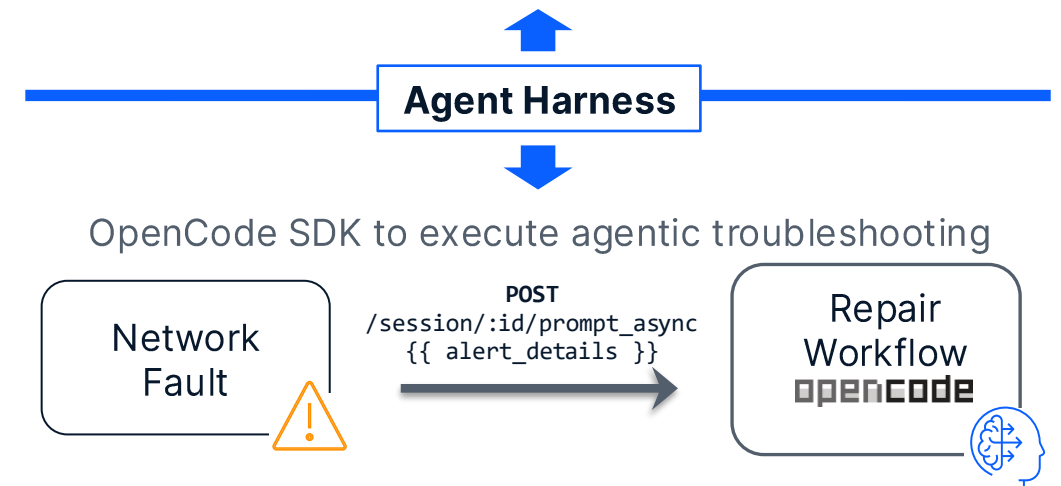
One harness for both sides: create fault intelligence as code *and* run repair workflows

Behavior lives in markdown files: edit agents, skills, and MCP config to iterate fast

Prototype now; port later to agent frameworks (e.g. LangChain) when scale demands it



OpenCode CLI to develop and test Intelligence Artifacts



Agents, Skills, and MCP for **developing** fault intelligence



IA Curator Agent

Create, optimize, and publish intelligence artifacts

Invoked by human engineer

Objectives

- Research source material for a fault condition
- Draft Fault Signatures and Repair Action Workflows
- Validate artifacts against schema, examples, and tests
- Publish artifacts for peer review through Git

Skills

ia-start

ia-research

ia-create

ia-optimize

ia-publish

Interfaces & Sources

Cisco Support API MCP

Cisco Docs MCP *

Git Repository

Schema Validator

Takeaways

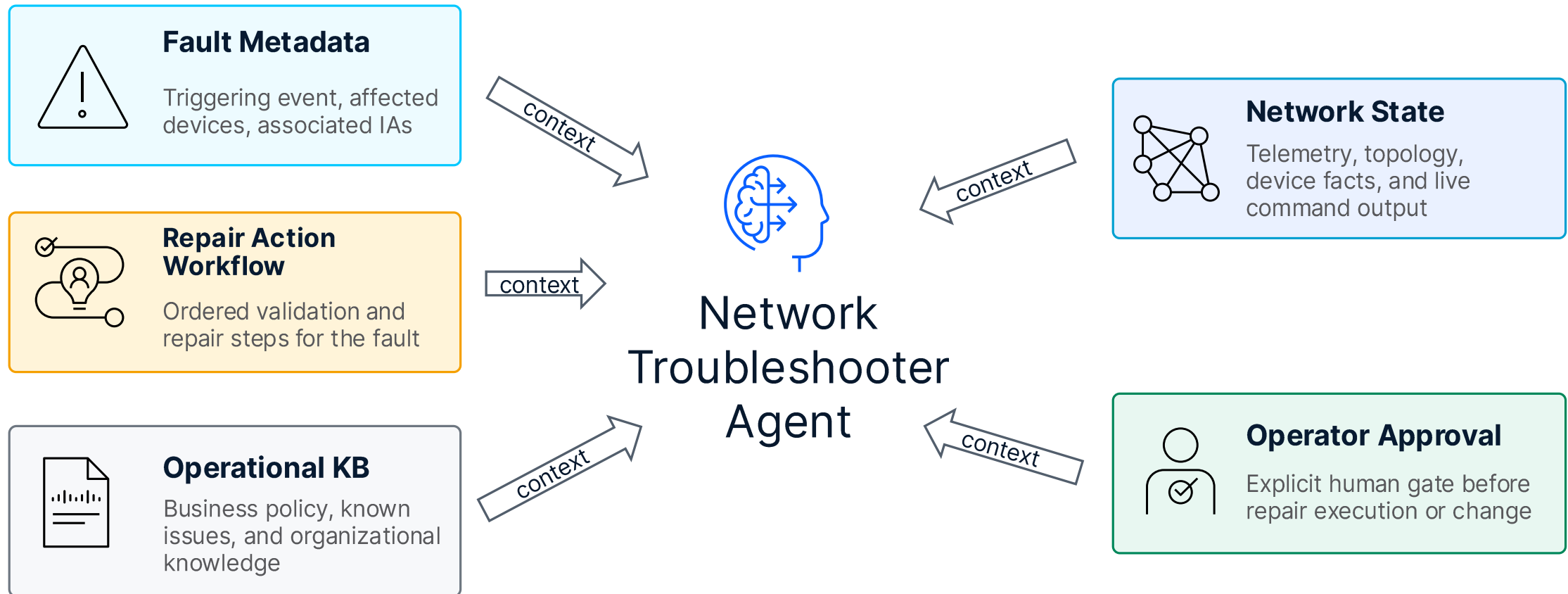
The agent is organized around an objective: turn source knowledge into curated artifacts

Skills define repeatable authoring procedures; interfaces connect the agent to MCP servers, repos, and validators

* Cisco Docs MCP planned for public release in 2H CY26

Context Engineering: Design for the “context bundle”

What context does the agent need to see to respond to a fault?



Agents, Skills, and MCP for operationalizing fault intelligence



Network Troubleshooter

Respond to a live fault using approved fault intelligence

Invoked by Splunk alert pipeline

Objectives

- Start auditable incident session
- Execute RAW with approval gates
- Resolve or escalate

Skills

fault-remediation

webex-notifier

Tools & External Systems

RADKit
MCP

Devices

Webex



IA Reader

Retrieve the matched intelligence artifact bundle

Invoked by network-troubleshooter

Objectives

- Find matched alert definition
- Return FS and RAW
- Keep artifact access read-only

Skills

None

Tools & External Systems

Git repository

Intelligence
Artifacts



KB Reader

Retrieve operational context for the active incident

Invoked by network-troubleshooter

Objectives

- Look up policy and windows
- Link to known issues
- Add business context

Skills

wiki-query

Tools & External Systems

Git repository

Markdown Wiki

Takeaways

- Runtime separates orchestration from context retrieval: the primary agent delegates to read-only specialists
- Safety boundary: runtime agents read repo/KB context; only approved interfaces touch operational systems

Agentic RAG for Knowledge Bases

Inspired by Andrej Karpathy's LLM Wiki approach: use agents to build knowledge that *compounds over time*

Traditional RAG over a chunked documents

VS.

Retrieval over compiled knowledge articles

Retrieval of vector data as fragments

What it does

- Chunk documents into vector store
- Retrieve top-N matches via semantic search
- Ask the LLM to synthesize

Agent-curated articles and links

How it works

- Read source material
- Extract durable facts and patterns
- Integrate into an editable wiki

Where it breaks down

Challenges with operationalizing

- Contextual document logic gets lost across chunks
- Complexity in operating and tuning
- More data sources can add noise

Benefits

Why it works

- Patterns compound over time
- Humans can edit and review
- Git preserves traceability

More info: <https://gist.github.com/karpathy/442a6bf555914893e9891c11519de94f>

Knowledge base design

Ingest new knowledge with agent harness, view in Obsidian, publish in Git



KB maintains operations policy, known issues, concepts, entities, etc

User manages via **kb-curator** agent

Runtime retrieval via **kb-reader** agent

Agents efficiently create and traverse relationships between articles via links

Skills for ingestion, structure, linting

- wiki
- wiki-ingest
- wiki-query
- wiki-lint
- save
- obsidian-markdown

> _templates

> wiki

> business-rules

> comparisons

> concepts

> entities

> incidents

> known-issues

> meta

> questions

> runbooks

> sources

hot

index

log

overview

AGENTS

README

SFP+ Silent Degradation — CRC Below BFD Threshold

Problem Description

SFP+ transceivers can fail gracefully — optical thresholds while signal integrity at the physical layer at a rate that is:

- High enough to cause quality degradation (e.g., latency-sensitive flows)
- Low enough that BFD does not trigger (BFD threshold)
- Low enough that BGP hold timers may not expire (BGP hold timer)

Result: The fault is invisible to standard monitoring tools and can persist indefinitely without a CRC-specific alert.

Related Incidents

Incident	Device	Details
INC-20260509 — Persistent CRC Errors xr-43	xr-43 TenGigE0/0/0/3	Local SFP+ failing; ~40 CRC/min; link UP; VoIP degradation; resolved by SFP swap
INC-20260314 — BGP Session Flap xr-43	xr-43 GigE0/0/0/1	Far-end (ISP-A PE) SFP degraded; CRC → BFD trigger → BGP flap; resolved by ISP-A

Knowledge automatically compiled for known issues spanning multiple incidents

Bringing it all together: Repository as Agent Harness

The repository establishes the control plane for all agentic operations

fault-intelligence-as-code/	
☐ .opencode/	agent + skill definitions
☐ agents/	network, reader, curator roles
☐ skills/	repeatable task procedures
☐ kb/	operational knowledge vault
☐ intelligence-artifacts/	published FS / RAW / RG bundles
☐ app/	FastAPI relay + approval bridge
☐ scripts/	simulator, tests, validation, indexes
☐ docs/	architecture + operations docs
☐ AGENTS.md	repo operating instructions
☐ opencode.json	models, MCP, per-agent allow-lists
Harness Config Knowledge Scripts and Services Docs	

How OpenCode uses the repository

1. Configure the Harness

Loads agent roles, skills, model selection, MCP endpoints, and per-agent tool permissions that define what each agent is allowed to do

2. Run Scripts and Services

The FastAPI alert relay starts OpenCode sessions from faults; the Webex approval bridge routes operator decisions back; helper scripts simulate and validate workflows

3. Supply Runtime Context

Reader agents retrieve approved intelligence artifacts and KB context so the Network Troubleshooter acts from versioned ground truth during execution.

Two approaches to agentic troubleshooting

Strict mode follows the approved workflow

Hybrid mode can investigate when the workflow is not enough



STRICT MODE

Ensures every action is grounded in **validated fault intelligence**

Follows the Repair Action Workflow exactly

Runs only predefined diagnostic and repair steps



HYBRID MODE

Provides **guided flexibility** when approved workflow does not fully cover scenario

Can gather additional evidence and reason outside the predefined steps

Keeps service-impacting changes approval-gated, and escalates when no safe action remains

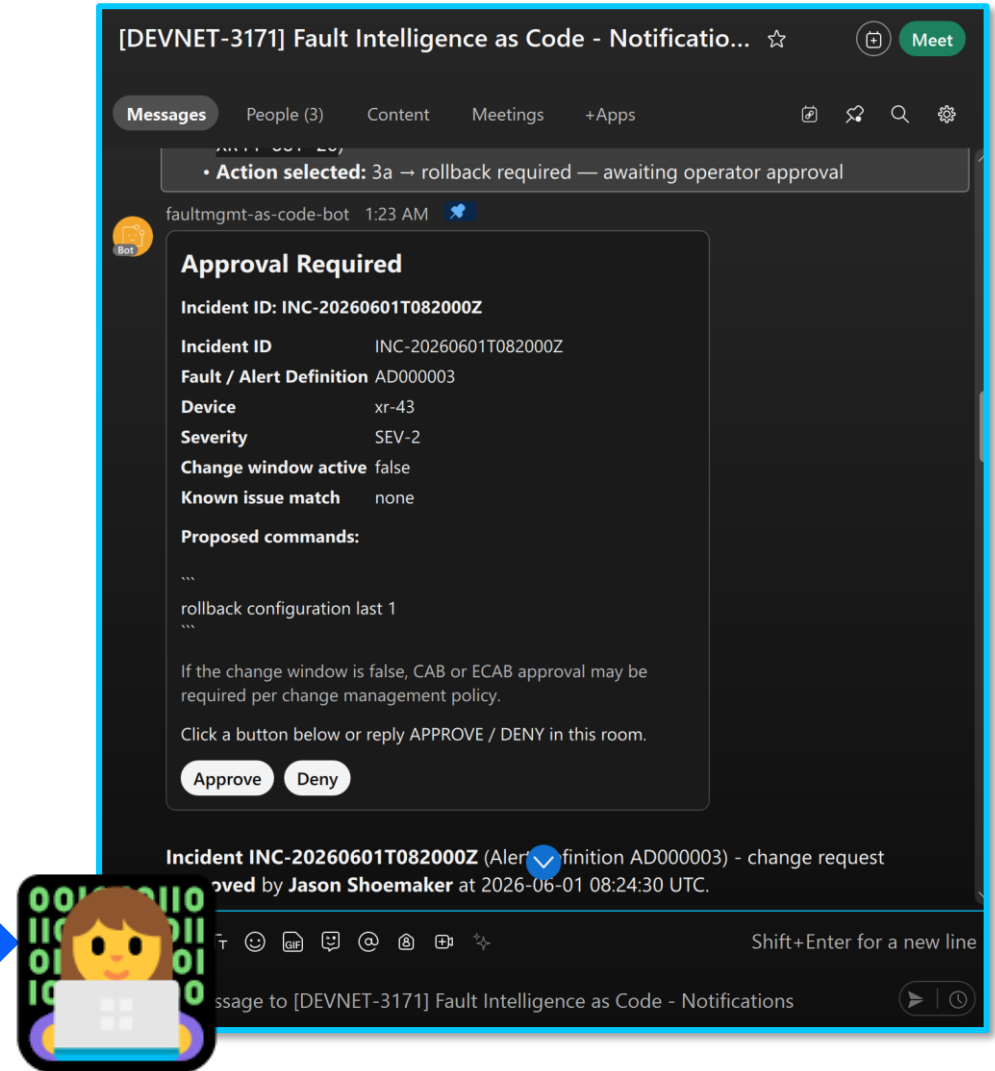
How do humans stay in the loop?

Layered policy across agent and skill governs RAW execution and approvals:

```
.opencode\skills\fault-remediation\SKILL.md  
.opencode\agents\network-troubleshooter.md
```

Policy: ***exec_cli** or **config_cli** actions MUST invoke the approval workflow*

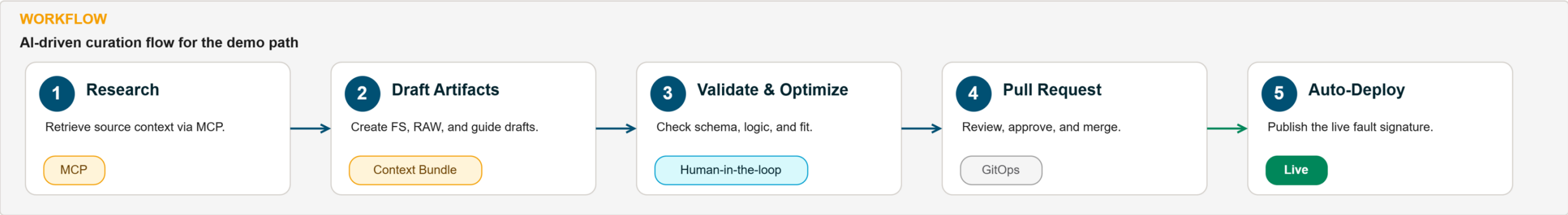
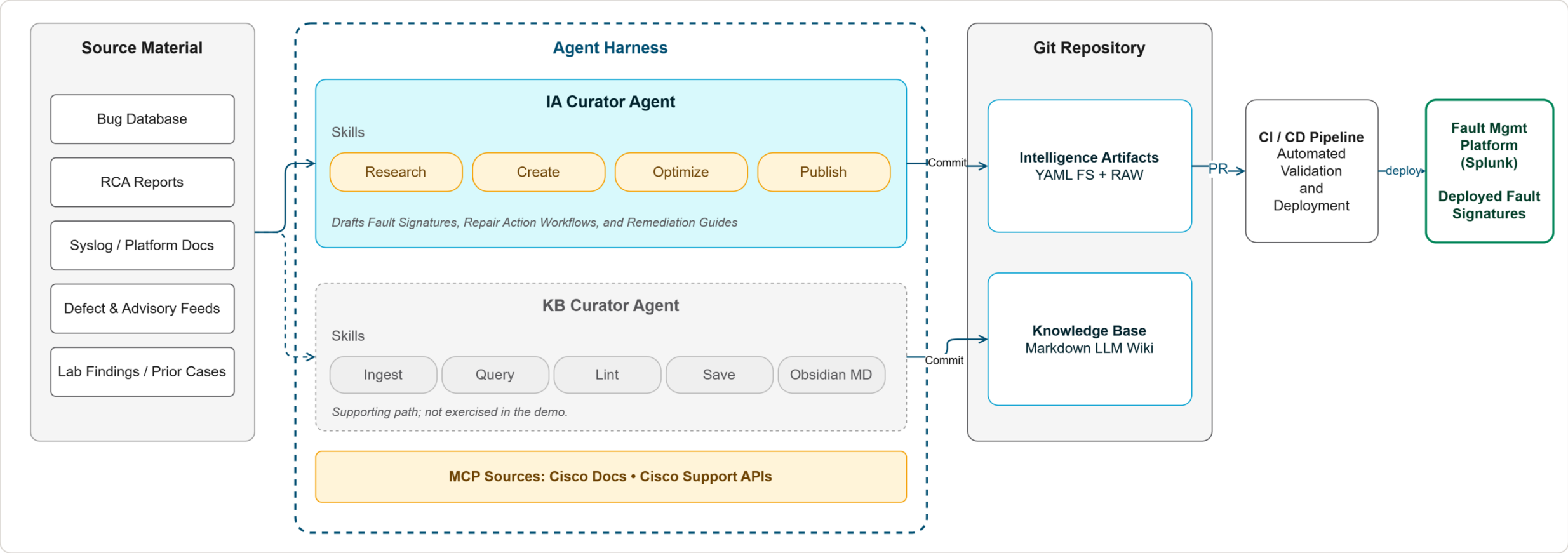
webex-notifier skill





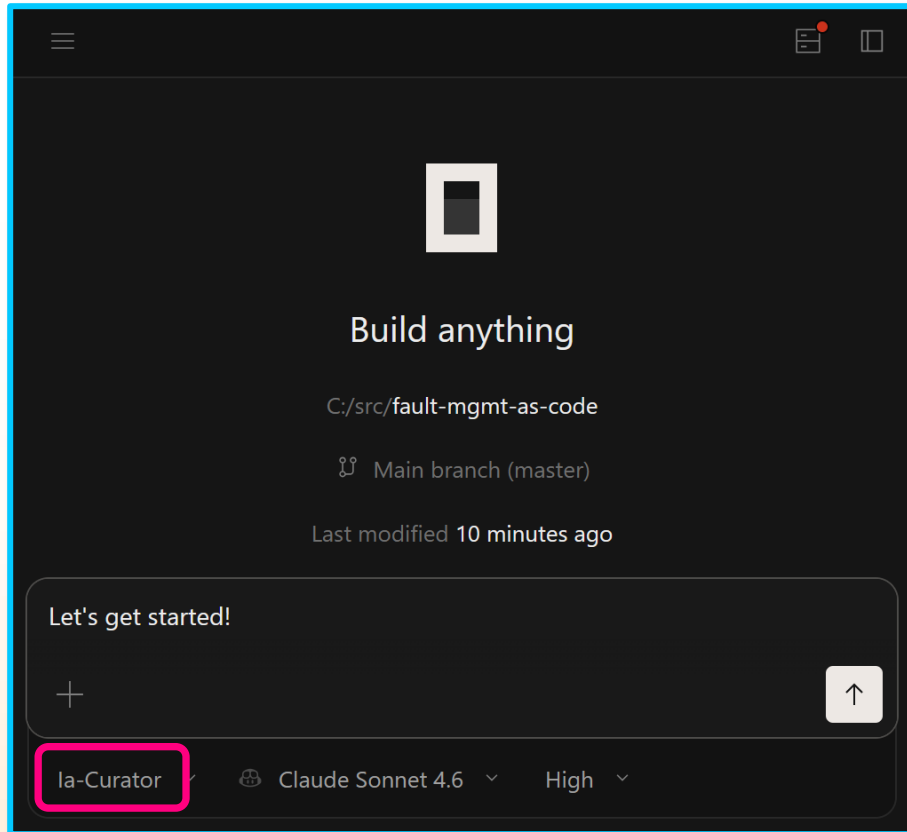
Fault Intelligence in Action

Demo: Compiling Fault Intelligence as Code

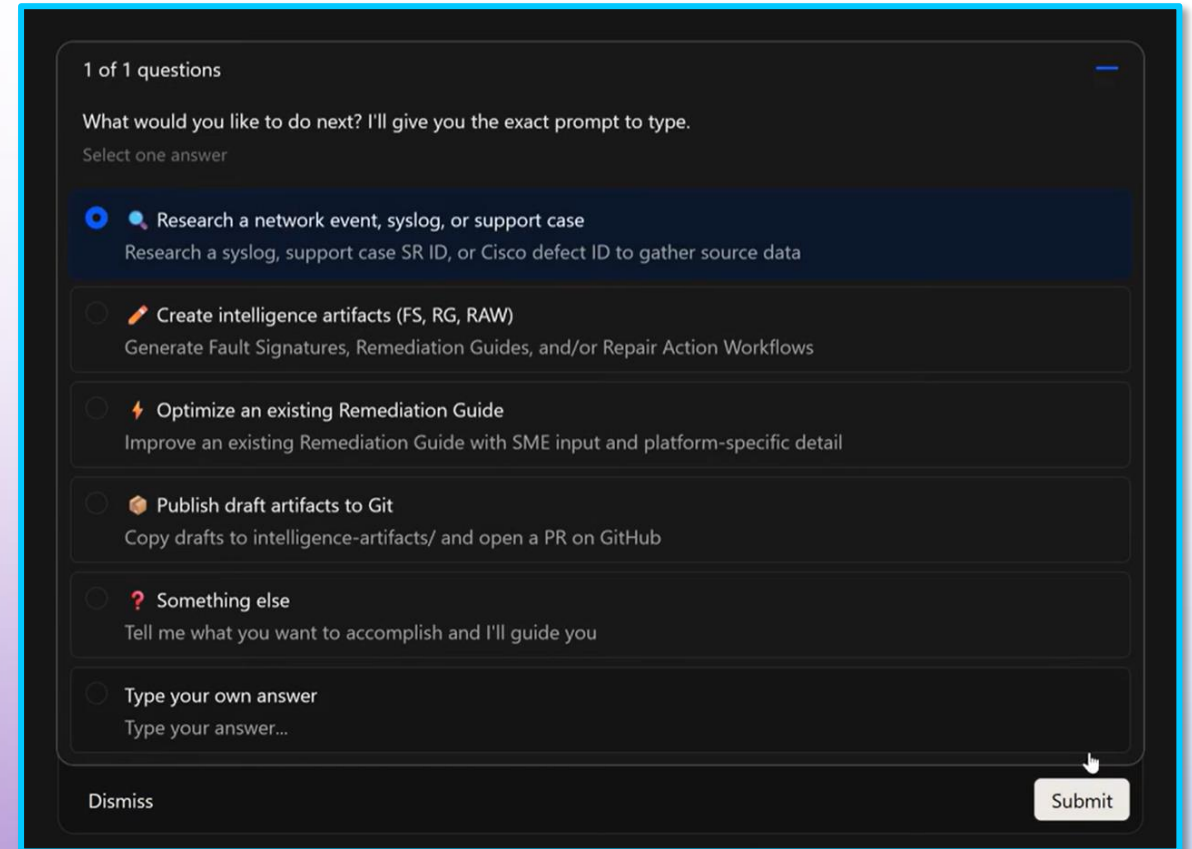


Getting Started

IA Curator and the ia-start skill



IA Curator agent



ia-start skill invoked

Researching issue detail to drive remediation intelligence

Research a Network Event, Syslog, or Support Case

Type one of the following in the chat box and press Enter:

ia-research SR 697123456

ia-research defect CSCwh12345

ia-research %FAN_TRAY-3-ABSENT

ia-research

The **ia-research** skill will query Cisco Support and Cisco Documentation in parallel, then save structured files to **research/<issue-name>/**. You review and edit those files before moving to artifact creation.

When ready to generate artifacts, type **ia-create** and the skill will pick up the research automatically.

Tip: Start a new chat (click the + button at the top of the Copilot Chat panel) before running **ia-research** to clear the context window and gives the next skill a clean slate to work with.

%ROUTING-BGP-5-ADJCHANGE : neighbor 172.20.20.17 Down - BGP Notification received, maximum number of prefixes reached (VRF: default) (AS: 65535)

la-Curator Claude Sonnet 4.6 High

Research complete. Files written to **research/bgp-max-prefix-adjchange-xr/**:

File	Contents
cisco-support-findings.md	3 defects analyzed — CSCtx30562 (key match, IOS XR, Sev 3), CSCur95359, CSCwd18821; consolidated syslog strings
cisco-docs-findings.md	Mnemonic explanation, both root cause paths, IOS XR config syntax, diagnosis CLI commands, software version notes
research-summary.md	Full RG recommendation table with all 8 sections mapped; coverage gap analysis; next steps

Results in human readable remediation guide

ia-create

+

↑

ia-Curator Claude Sonnet 4.6 High

Remediation Guide: BGP Neighbor Down — Neighbor's Maximum Prefix Limit Exceeded by Local Advertisement

Alert Definition: AD000000 Guide ID: RG000000 Linked Fault Signature: FS000000
Linked Repair Action Workflow: RAW000000

Overview

This guide diagnoses and resolves BGP session failures on IOS XR routers where a BGP neighbor sends a CEASE notification to the local router because the local router is advertising more prefixes than the neighbor's configured inbound maximum-prefix limit allows. The local router logs `%ROUTING-BGP-5-ADJCHANGE : neighbor <ip> Down - BGP Notification received, maximum number of prefixes reached` and the session drops. The keyword **"BGP Notification received"** is the defining indicator: the neighbor initiated the teardown, not the local router. The root cause is on the local router's outbound advertisement policy, and the fix is to reduce what is being advertised to that neighbor — via an outbound route policy or prefix filter — or to coordinate with the neighbor to raise their limit, then restore the session.

Applicability

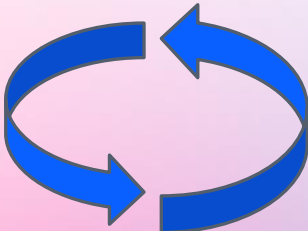
- **Products:** Cisco ASR 9000 Series Aggregation Services Routers; Cisco 8000 Series Routers; Cisco NCS 5000, 5500, 5700 Series; Cisco NCS 540, 560 Series; all other IOS XR platforms sharing the BGP routing process
- **Operating Systems:** IOS XR (all releases)
- **Component:** Route Processor
- **Severity:** Warning
- **Related Defects:** None

Triggering Events

Event 1: BGP Neighbor Down — Neighbor's Inbound Maximum Prefix Limit Reached

- **Type:** Syslog
- **Message ID:** ROUTING-BGP-5-ADJCHANGE
- **Example Message:**

```
bgp[1090]: %ROUTING-BGP-5-ADJCHANGE : neighbor 172.20.20.17 Down - BGP Notification received, maximum number of prefixes reached (VRF: default) (AS: 65535)
```



ia-optimize

+

↑

ia-Curator Claude Sonnet 4.6 High

Clarifying Questions

The following questions must be answered before the optimized Remediation Guide can be generated. Replace `<your answer here>` in each **Answer** block. Partial answers are accepted — unanswered questions will become `[GAP]` placeholders in the guide.

Undefined Thresholds

T1. Step 2 states to check "commits made within the last few hours" and suggests inspecting "last 5" commits if the most recent shows no change. What is the appropriate lookback window (in time or number of commits) before the engineer concludes no relevant config change exists and escalates?

Answer: Use a simple operational default: review commits from the last **4 hours** and inspect up to the **last 5 commits** first.

T2. Step 4 states the session should recover "within 30–120 seconds" after `clear bgp`. At what specific point (elapsed time or number of BGP open attempts) should the engineer conclude the session is not recovering and escalate?

Answer: Use **120 seconds** as the main threshold. After `clear bgp`, if the session is still not **Established** within **2 minutes**, or if it clearly fails through **2–3 open attempts**, stop waiting and escalate.

Missing Decision Branches

D1. Step 2 Decision Point: if no route-policy or prefix-set change is found in the commit history, the guide immediately escalates. Should there be an intermediate check before escalating — for example, verifying whether the prefix count grew due to a new peer being established or a redistribution policy change rather than a local route-policy edit? Or is "no config change found → escalate immediately" the intended path?

Answer: Keep this simple. Do **one quick sanity check** before escalating: confirm which export route-policy / prefix-set is attached to the neighbor and confirm there was no obvious recent redistribution or policy change affecting that neighbor. If there is still no clear recent config change, **escalate**. Do **not** expand this guide into deep `advertised-routes` troubleshooting.

D2. Step 3 Decision Point: the rollback preview (`show configuration rollback changes last 1`) might show a policy that is already identical to the current running config — meaning the commit that was found in Step 2 changed something else but the

Remediation guide then drives intelligent artifact creation

ia-create usage

1 of 1 questions

Which artifact types would you like to generate from this Remediation Guide?
Select all answers that apply

☒ **Fault Signature (FS)**
Machine-readable syslog/alarm detection pattern (YAML). Syslog patterns detected in the RG's Triggering Events section.

☒ **Repair Action Workflow (RAW)**
Automated step-by-step repair workflow (YAML). Branching diagnosis and repair actions detected in the RG.

```
✓ AD000003-bgp-max-prefix-adjchange-xr
  ✓ docs
    ↓ FS000003.analysis.md
    ↓ RAW000003.analysis.md
    ↓ RG000003-bgp-max-prefix-adjchange-questionnaire.md
    ↓ RG000003.analysis.md
  ✓ tests
    ! RAW000003-BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED_REPAIR.tests.yml
    ! FS000003-BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED.yml
    ! RAW000003-BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED_REPAIR.yml
    ↓ RG000003-bgp-max-prefix-adjchange-guide.md
```

Example of a Fault Signature IA for our scenario

```
schema-version: "0.0.1"

metadata:
  name: BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED
  id: "FS000003"
  alert_def_id: "AD000003"
  version: "1.0.0"
  description: |
    Detects an IOS XR BGP neighbor session that is torn down by the remote peer
    because the local router advertised more prefixes than the peer's configured
    inbound maximum-prefix limit allows. The signature matches the
    ROUTING-BGP-5-ADJCHANGE syslog containing "BGP Notification received,
    maximum number of prefixes reached" and extracts the neighbor IP address,
    VRF name, and remote AS number for use by the paired Repair Action Workflow.
  severity: WARNING
  created_date: "2026-05-29"
  modified_date: "2026-05-29"
  author: "CX Intelligence Team"
  product_ids:
    - "ASR 9000.*"
    - "Cisco 8000.*"
    - "NCS 5000.*"
    - "NCS 5500.*"
    - "NCS 5700.*"
    - "NCS 540.*"
    - "NCS 560.*"
  os_versions:
    - "IOS.XR .*"
  component: BGP
  tags:
    - bgp
    - routing
    - max-prefix
    - ios-xr
    - cisco-live-us-2026
    - devnet-3171
  priority: 5
  symptom: "LINK_DOWN"
  repair_action_workflow_ref: BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED_REPAIR
```

```

conditions:
  logic: "E1"
  logic_lookback_time: 0
  events:
    - event:
        id: "E1"
        type: syslog
        path: "syslog"
        message_type: "ROUTING-BGP-5-ADJCHANGE"
        message_sample: "bgp[1090]: %ROUTING-BGP-5-ADJCHANGE : neighbor 172.20.20.17 Down - BGP Notification r
        evaluation:
            type: regex
            value: "%ROUTING-BGP-5-ADJCHANGE\\s*:\\s*neighbor\\s+(\\S+)\\s+Down\\s+-\\s+BGP Notification r
            parameters:
                - type: extract_to_variable
                  name: neighbor_ip
                  source: "match.group(1)"
                  description: "IP address of the affected BGP neighbor"
                - type: extract_to_variable
                  name: vrf_name
                  source: "match.group(2)"
                  description: "VRF in which the BGP neighbor session exists"
                - type: extract_to_variable
                  name: neighbor_as
                  source: "match.group(3)"
                  description: "Remote AS number reported in the syslog"
            match_count: 1
            match_period: 0
            clear_event:
                pattern: "%ROUTING-BGP-5-ADJCHANGE\\s*:\\s*neighbor\\s+{{ neighbor_ip }}\\s+Up\\s+\\s+\\s+(VRF:\\s*{
                lookback_period: 300
                correlation_variables:
                    - neighbor_ip
                    - vrf_name
                    - neighbor_as

```

Example of a Repair Action Workflow IA for our scenario

```
workflow:
  metadata:
    name: BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED_REPAIR
    id: "RAW000003"
    alert_def_id: "AD000003"
    version: "1.0.0"
    description: |
      Diagnoses and repairs an IOS XR BGP neighbor session that was torn down by
      the remote peer after the local router exceeded the peer's inbound
      maximum-prefix limit. The workflow confirms the fault from BGP neighbor
      state, identifies the attached export policy, verifies that the most recent
      commit is a safe rollback candidate, rolls back the latest commit, clears
      the BGP session, and verifies recovery. If the rollback path is unclear or
      the session does not recover, the workflow escalates with evidence.

  inputs:
    - name: neighbor_ip
      source: "{{ alert_vars.neighbor_ip }}"
      type: string
      description: "IP address of the affected BGP neighbor extracted by FS000003"
    - name: vrf_name
      source: "{{ alert_vars.vrf_name }}"
      type: string
      description: "VRF name extracted by FS000003"
    - name: neighbor_as
      source: "{{ alert_vars.neighbor_as }}"
      type: string
      description: "Remote AS number extracted by FS000003"
```

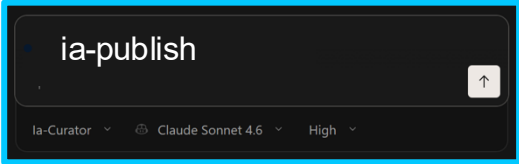
```
steps:
  - step_id: "1"
    name: confirm_max_prefix_cease
    description: |
      Confirm that the BGP session is down due to a neighbor-sent max-prefix
      CEASE notification, not a different adjacency failure.
    validation:
      and:
        - eval_cli:
            commands:
              - "show bgp neighbors {{ neighbor_ip }} | include \"BGP state\""
            pattern: "BGP state = (Idle|Active)"
            outputs:
              - name: session_down
                source: "{{ result.matched }}"
                type: boolean
                description: "True when the neighbor is not in Established state"
        - eval_cli:
            commands:
              - "show bgp neighbors {{ neighbor_ip }} | include \"Last reset\""
            pattern: "Maximum Number of Prefixes Reached"
            outputs:
              - name: max_prefix_cease_detected
                source: "{{ result.matched }}"
                type: boolean
                description: "True when the last reset reason is a max-prefix CEASE"
    action_select:
      - action_id: "1a"
        name: fault_confirmed
        description: "Session is down and the reset reason matches the expected max-prefix CEASE."
        conditions:
          - condition: "session_down == True and max_prefix_cease_detected == True"
        action_groups:
          - actions:
              - goto:
                  step_id: "2"
      - action_id: "1b"
        name: different_fault
        description: "The session is not down for the expected max-prefix reason."
        conditions:
          - condition: "default"
        action_groups:
          - actions:
              - escalate:
```

Example of test artifacts created

```
# Test bundle for RAW000003 BGP Neighbor Max Prefix Limit Exceeded Repair.
#
# Diagnoses and repairs an IOS XR BGP neighbor session that was torn down by the
# remote peer after the local router exceeded the peer's inbound maximum-prefix
# limit. Confirms the max-prefix CEASE condition, identifies the neighbor's
# outbound route policy, verifies the latest commit and rollback preview are
# relevant, rolls back the most recent commit, clears the BGP session, and
# verifies recovery.
#
# Terminal leaves covered here:
#
# 1. 1b → escalation (session is not down for th
# 2. 1a→2b → escalation (outbound route policy not
# 3. 1a→2a→3b → escalation (latest commit is not a cle
# 4. 1a→2a→3a→4b → escalation (rollback preview does not
# 5. 1a→2a→3a→4a→5a→6a → resolution (default VRF rollback + cle
# 6. 1a→2a→3a→4a→5b→6a → resolution (non-default VRF rollback +
# 7. deny-approval at Step 4 → escalation (agent-only rollback denial
#
# 8. 1a→2a→3a→4a→5a→6b→7a → escalation (same max-prefix CEASE recu
# 9. 1a→2a→3a→4a→5a→6b→7b → escalation (different post-rollback is
# 10. 1a→2a→3a→4a→5b→6b→7a → escalation (same max-prefix CEASE recu
# 11. 1a→2a→3a→4a→5b→6b→7b → escalation (different post-rollback is
#
```

```
tests:
- name: escalate_different_fault_cause
  source: synthetic
  description: |
    The neighbor is down, but the last reset reason is not the expected
    max-prefix CEASE. This RAW should stop immediately and escalate because
    the rollback path does not apply.
  expected:
    outcome: "escalation"
    step_path: ["1"]
    action_ids: ["1b"]
    variables:
      session_down: true
      max_prefix_cease_detected: false
  alert_payload:
    mode: strict
    alert_def_id: "AD000003"
    device_hostname: "xr-edge-01"
    alert_vars:
      neighbor_ip: "192.0.2.17"
      vrf_name: "default"
      neighbor_as: "64513"
  responses:
    - step_id: "1"
      command: "show bgp neighbors 192.0.2.17 | include \"BGP state\""
      output: "BGP state = Active"
    - step_id: "1"
      command: "show bgp neighbors 192.0.2.17 | include \"Last reset\""
      output: "Last reset 00:00:39 ago, due to Hold Timer Expired"
  approvals:
    default: "APPROVED"
```

Push up to repo after review and testing



1 of 1 questions

Select the artifact files to publish. All are pre-selected — deselect any to exclude.
Select all answers that apply

☒

AD000000-bgp-max-prefix-adjchange-xr / FS000000-BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED (Recommended)
Fault Signature draft with placeholder IDs; companion analysis will be copied automatically.

☒

AD000000-bgp-max-prefix-adjchange-xr / RG000000-bgp-max-prefix-adjchange-guide.md — RG (Recommended)
Optimized Remediation Guide markdown draft; companion docs will be copied automatically.

☒

AD000000-bgp-max-prefix-adjchange-xr / RAW000000-BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED_REPAIR.yml — RAW (Recommended)
Repair Action Workflow draft with placeholder IDs; companion analysis will be copied automatically.

Issue	✓ Created	#7 — https://github.com/ciscomanagedservices/fault-mgmt-as-code/issues/7
Branch	✓ Created	feat/bgp-max-prefix-adjchange-xr
Commit	✓ Committed	160b5d9 feat: add BGP max-prefix intelligence artifacts
Push	✓ Pushed	origin/feat/bgp-max-prefix-adjchange-xr
PR	✓ Opened	#8 — https://github.com/ciscomanagedservices/fault-mgmt-as-code/pull/8
Index PR link backfill	✓	11d7270 pushed with PR link in index.md

IA changes are peer reviewed and merged into production

Add BGP max-prefix intelligence artifacts for IOS XR Live US 2026 Demo] #8

Open

jashoema wants to merge 2 commits into master from feat/bgp-max-prefix-adjchange-xr

Conversation 0

Commits 2

Checks 1

Files changed 10

jashoema commented 2 minutes ago

...

Resolves #7

Summary

Adds a linked intelligence-artifact set for IOS XR BGP max-prefix adjchange handling. This PR publishes the new Fault Signature, Markdown Remediation Guide, and Repair Action Workflow for the case where a remote BGP peer sends a CEASE notification because the local router exceeded the peer's inbound maximum-prefix limit.

Field	Value
Customer	Cisco Live US 2026 Demo
Engagement	DEVNET-3171 — Fault Intelligence as Code

Artifact Changes

Action	Type	ID	Name
Added	Fault Signature	FS000003	BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED
Added	Remediation Guide	RG000003	BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED_GUIDE
Added	Repair Action Workflow	RAW000003	BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED_REPAIR

Open Design Questions

FS000003

1. Regex stability: Confirm the exact phrase maximum number of prefixes reached is stable across all target IOS XR releases and platforms.

2. Prefix-set attribution: This signature detects the fault from syslog only; it does not identify which

docs

index.html

intelligence-artifacts

AD000003-bgp-max-prefix-adjch...

docs

FS000003.analysis.md

RAW000003.analysis.md

RG000003-bgp-max-prefix-ad...

RG000003.analysis.md

FS000003-BGP_NEIGHBOR_MA...

RAW000003-BGP_NEIGHBOR_...

RG000003-bgp-max-prefix-adjc...

index.json

index.md

Friendly UI for navigating the fault intelligence artifact base

INTELLIGENCE ARTIFACT EXPLORER

3 FS 4 RG 3 RAW 4 groups 11 platforms

Paste a syslog message to match against signatures...

Syslog regex match

Keyword search

Search by ID

All Platforms

All Components

All Severities

FS

RG

RAW

Showing 10 of 13 artifacts

AD000001-bgp-prefix-anomaly

3 artifacts

FS

BGP_PREFIX_ANOMALY

WARNING · ROUTE_PROCESSOR · NCS-55A1-36H-SE-S, 8808-SYS, 8818-SYS

FS000001 v1.0.0

RAW

BGP_PREFIX_ANOMALY_REPAIR

RAW000001 v1.0.0

RG

Remediation Guide: BGP Prefix Anomaly

RG000001 v

AD000002-bgp-neighbor-admin-shutdown-xr

3 artifacts

FS

BGP_NEIGHBOR_ADMIN_SHUTDOWN

WARNING · CONTROLLER · ASR 9000*, Cisco 8000*, NCS 5[57]00*, NCS 540*

FS000002 v1.0.0

RAW

BGP_NEIGHBOR_ADMIN_SHUTDOWN_REPAIR

RAW000002 v1.0.0

RG

Remediation Guide: BGP Neighbor Administrative Shutdown

RG000002 v

AD000003-bgp-max-prefix-adjchange-xr

3 artifacts

FS

BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED

WARNING · BGP · ASR 9000*, Cisco 8000*, NCS 5000*, NCS 5500*...

FS000003 v1.0.0

RAW

BGP_NEIGHBOR_MAX_PREFIX_LIMIT_EXCEEDED_REPAIR

RAW000003 v1.0.0

RG

Remediation Guide: BGP Neighbor Down - Neighbor's Maximum Prefix Limit Exceeded by Local Adverti...

RG000003 v

docs

1 artifact

RG

Optimization Questionnaire: BGP Neighbor Down - Neighbor's Maximum Prefix Limit Exceeded by Local Adver...

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DEVNET-3171

43

Which can drive automated provisioning to your NMS



✓ All checks have passed
1 successful check

✓  Test / test (pull_request) Successful in 6s

✓ No conflicts with base branch
Merging can be performed automatically.

Merge pull request

You can also merge this with



All checks have passed
1 successful check

✓  Deploy / deploy (push) Successful in 7s



Settings

Alert ad000003_bgp_neighbor_max_prefix_limit_exceeded

Description Detects an IOS XR BGP neighbor session that is torn down by the remote peer because the local router advertised more prefixes than the peer's configured

Search

```
index=syslog sourcetype=cisco:ios "ROUTING-BGP-5-ADJCHANGE"
| regex _raw="%ROUTING-BGP-5-ADJCHANGE\\s*:\\s*neighbor\\s+(\\S+)\\s+Down\\s+\\s
+BGP Notification received,\\s+maximum number of prefixes reached\\s+\\(VRF
:\\s*(\\S+)\\)\\s+\\(AS:\\s*(\\d+)\\)"
| rex field=_raw "%ROUTING-BGP-5-ADJCHANGE\\s*:\\s*neighbor\\s+(?<neighbor_ip>\\S+)\\s
+Down\\s+\\s+BGP Notification received,\\s+maximum number of prefixes reached\\s
+\\(VRF:\\s*(?<vrf_name>\\S+)\\)\\s+\\(AS:\\s*(?<neighbor_as>\\d+)\\)"
| eval alert_def_id="AD000003"
| table _time device_name device_ip alert_def_id neighbor_ip vrf_name neighbor_as
_raw
```

Trigger Actions

+ Add Actions

When triggered

>  Add to Triggered Alerts Remove

✓  Webhook Remove

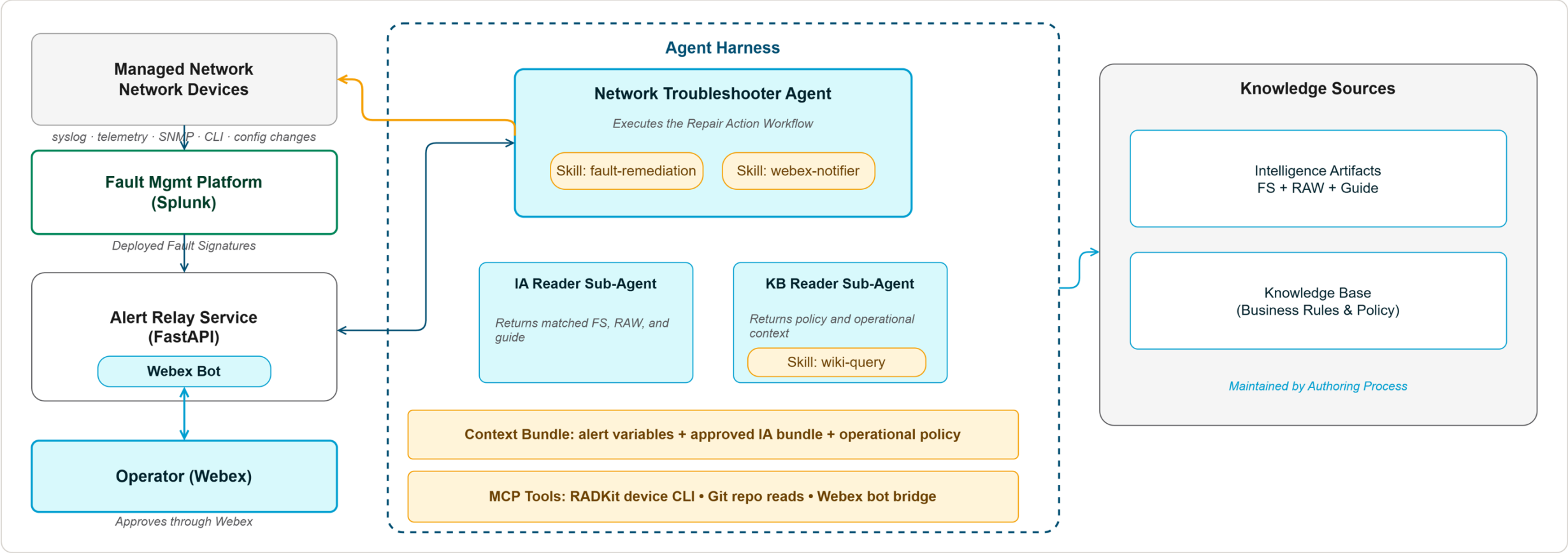
URL

Specified URL to send JSON payload via HTTP POST (ex., https://your.server.com/api/v1/webhook).

[Learn More](#)



Demo: Agentic Troubleshooting of Live Faults



WORKFLOW

Runtime fault response flow



Fault intelligence in action: detection & mitigation

The screenshot displays a web interface for fault intelligence, divided into two main panels. A large blue arrow points from the left panel to the right panel.

Left Panel (Session Details):

- Session:** BGP neighbor down: max prefixes exceeded on xr-43
- Changes:** Proposed command: `rollback configuration last 1`
- Approval context:**
 - Objective: Roll back the most recent IOS XR config commit that attached route-policy XR43-TO-XR44-OUT-20 to neighbor 172.20.20.18, causing the max-prefix CEASE notification
 - Key evidence: BGP state=Idle; outgoing policy=XR43-TO-XR44-OUT-20; latest commit sets this policy on neighbor 172.20.20.18 ipv4 unicast
 - Risk/impact: Rollback removes last commit; outbound route advertisement to AS 3334 will revert to prior policy; BGP session will be cleared in the next step
 - Expected result: Config rolled back; session will be cleared and should recover to Established
 - Verification plan: Step 5 will confirm BGP state = Established after session clear
- Warning:** ⚠ No active change window. Change management policy requires approval.
- Status:** Awaiting the human operator response: APPROVED or DENIED

Right Panel (Notifications):

[DEVNET-3171] Fault Intelligence as Code - Notifications

Messages:

- attached route-policy XR43-TO-XR44-OUT-20 to neighbor 172.20.20.18, causing max-prefix CEASE
- Key evidence:** BGP state=Idle; outgoing policy=XR43-TO-XR44-OUT-20; latest commit sets this policy on neighbor 172.20.20.18 ipv4 unicast
- Proposed commands:**

```
...
rollback configuration last 1
...
```
- Risk / impact:** Rollback removes last commit; outbound route advertisement to AS 3334 will revert to prior policy; BGP session will be cleared in next step
- Expected result:** Config commit rolled back; BGP session will be cleared and should recover to Established
- Verification:** Step 5 will confirm BGP state = Established after session clear
- If the change window is **false**, CAB or ECAB approval may be required per change management policy. Full evidence is in the preceding step-progress messages.
- Click a button below or reply **APPROVE / DENY** in this room.
- Buttons:** Approve Deny

Seen by: [User Icon]

Footer: Write a message to [DEVNET-3171] Fault Intelligence as Code - Notifications

Lessons learned from customers

- Operator + Network Vendor + Unified Data Models = Automated Support Cases
- Enable “Closed Loop RCA” with agents writing to GitHub Issues
- Most IAs tend to be for diagnosing faults, isolating devices, and escalating
- Hybrid mode also matters – explore always-on passive fault diagnosis
- Use GenAI to quickly integrate IA schemas with your existing tools
- Reduce toil – give your network vendors an IA schema for their deliverables!

Now you've seen it working...



Knowledge of data models for operational fault management



See how to integrate fault intelligence-as-code with agentic operations

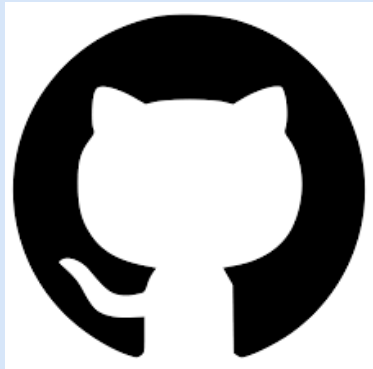


Open-sourced code that you can run for your operations

Interested in applying Fault as Code principles to your operations?

Do it yourself

Pull down our code from github and test in your lab



<https://cs.co/fault-intel>

Let Cisco services help you

Engage with Cisco Customer Experience to help you implement digital fault signatures and repair with **Cisco Professional Services**.



And now you have everything you need to start practicing this discipline!



Knowledge of data models for operational fault management



See how to integrate fault intelligence-as-code with agentic operations



Open-sourced code that you can run for your operations

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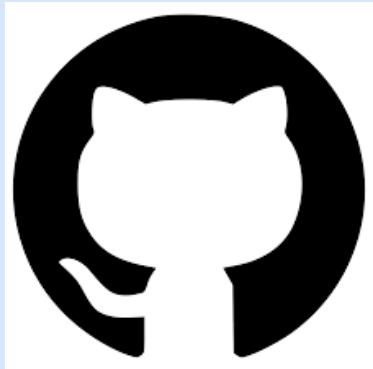


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Questions?

Do it yourself

Pull down our code from github and test in your lab



<https://cs.co/fault-intel>

Let Cisco services help you

Engage with Cisco Customer Experience to help you implement digital fault signatures and repair with Cisco Professional Services.



CISCO Live !

Thank you



